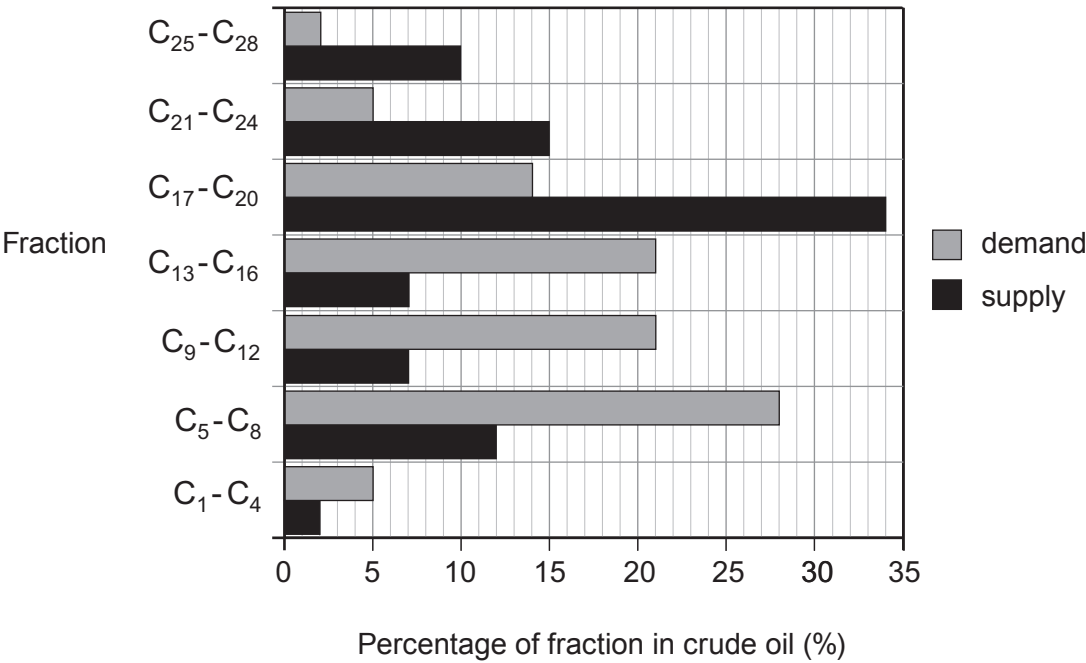


3. (a) The bar chart shows the relative *supply* and *demand* for some fractions obtained from crude oil.



(i) Use the bar chart to describe how the **difference** between *supply* and *demand* of the fractions changes as chain length increases. [2]

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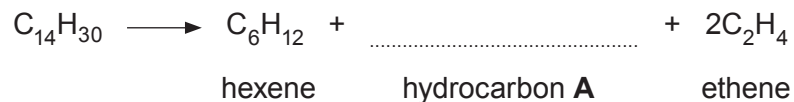
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- (ii) Oil companies have solved the problem of the over-supply of some fractions by using a process called cracking.

$C_{14}H_{30}$ can be cracked forming hexene, ethene and hydrocarbon **A**.

- I. Complete the equation for the cracking of $C_{14}H_{30}$. [1]



- II. Name hydrocarbon **A**. [1]

- III. State why ethene is considered an important raw material. [1]

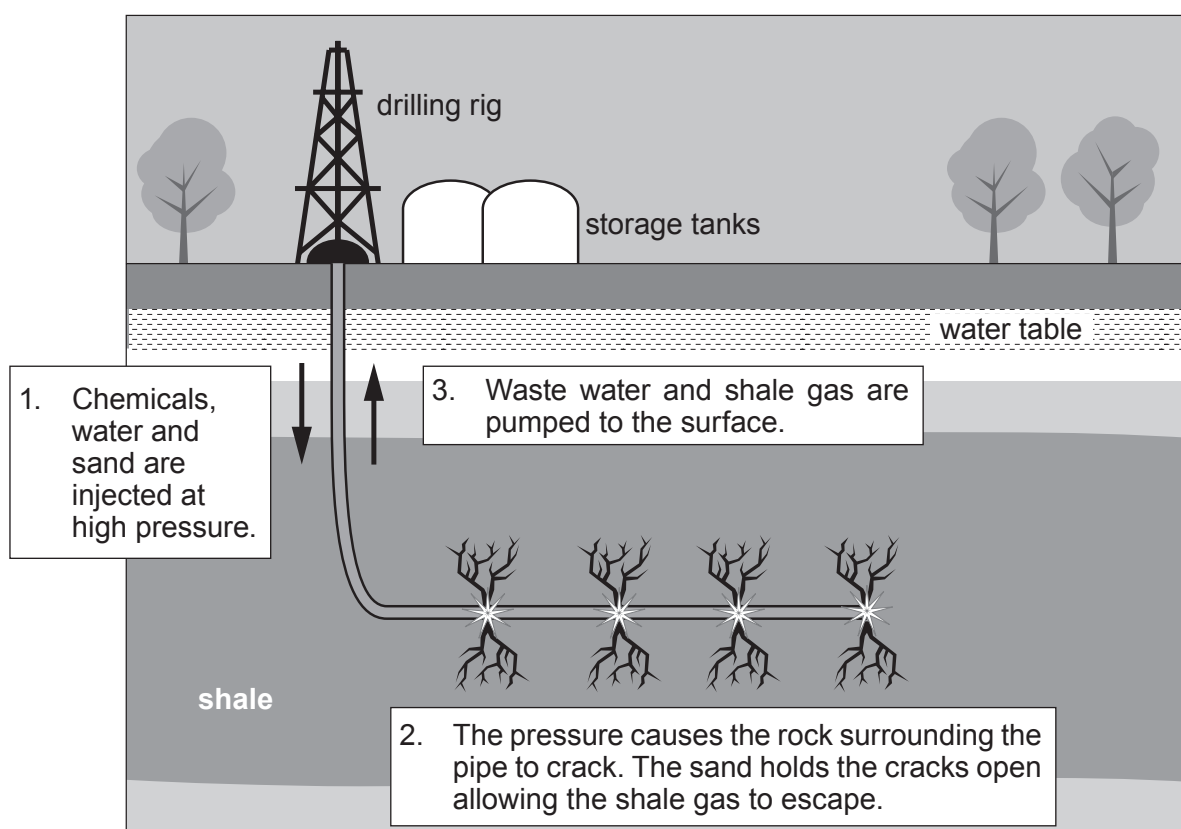
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- (iii) One hydrocarbon found in the C_1 - C_4 fraction is propane. Propane burns in air forming carbon dioxide and water.

Balance the symbol equation that represents this reaction. [1]



- (b) Shale gas is natural gas trapped in rocks deep underground. Like oil and coal, shale gas has, essentially, formed from the remains of plants, animals and micro-organisms that lived millions of years ago. It can be extracted using a process known as hydraulic fracturing – or “fracking” – which involves drilling long horizontal wells into the rocks more than a kilometre below the surface. Massive quantities of water, sand and chemicals are pumped into the wells at high pressure. This opens up cracks in the shale, which are held open by the sand, enabling the trapped gas to escape to the surface for collection.



Supporters of fracking argue that extracting shale gas deposits will help keep energy affordable and cut consumption of dirtier coal. But opponents claim fracking is dangerous and polluting, and that tapping into extra shale gas supplies will increase carbon dioxide emissions. The main controversy surrounding shale gas is the potential of fracking to contaminate drinking water supplies with shale gas or drilling chemicals. Other issues include the huge quantities of water and chemicals used in the extraction process, the waste water generated and possible earthquake tremors.



(i) Put a tick (✓) in the box next to the statement that identifies the substance(s) recovered during the fracking process. [1]

only shale gas

☐

shale gas and contaminated water

☐

shale gas and sand

☐

shale gas, sand and toxic chemicals

☐

(ii) Put a tick (✓) in the box next to the statement which is true. [1]

burning shale gas does not cause global warming

☐

chemicals used in fracking are contaminating our drinking water

☐

fracking produces vast quantities of contaminated water

☐

shale gas is a renewable energy source

☐

shale gas is cheaper than other fossil fuels

☐

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3. The tables show the molecular formulae of some alkanes and alkenes.

Alkanes	Alkenes
CH_4	C_2H_4
C_2H_6	C_3H_6
C_3H_8	
C_4H_{10}	

- (a) The general formula for the alkene family is C_nH_{2n} . Give the general formula for the alkane family. [1]

.....

- (b) When alkanes and alkenes completely burn in air they form the same two products.

Give the chemical formulae for both products. [1]

..... and

- (c) Draw the structural formula for propene. [1]

- (d) Bromine water is used to distinguish alkenes from alkanes. Describe the colour **change** seen when bromine water is added to an alkene. [1]

.....



3. (a) The list below shows part of the reactivity series.

sodium
aluminium
(carbon)
tin
copper
silver

- (i) Tin is extracted from its ore by heating with carbon. Aluminium is extracted from its ore using a different method. Give the name of the method used to extract aluminium. [1]

.....

- (ii) The equation shows the extraction of tin from tin oxide using carbon.



Tick (✓) the box next to the correct statement. [1]

Carbon is reduced

☐

Tin is oxidised

☐

Tin oxide is reduced

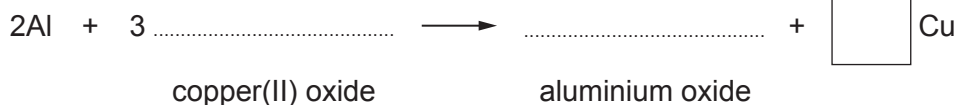
☐

Carbon dioxide is oxidised

☐

- (iii) When aluminium and copper(II) oxide are heated together, aluminium oxide and copper are formed.

Complete and balance the equation for this reaction. [3]



(b) A teacher wanted to find out the position of four metals **A**, **B**, **C** and **D** in the reactivity series.

She heated each metal in turn with oxides of the other three. The results were as follows.

A reduced the oxide of **C**

B reduced the oxide of **A**

B reduced the oxide of **C**

D reduced the oxide of **B**

Place the metals in order of reactivity. [2]

Most reactive

.....

.....

Least reactive

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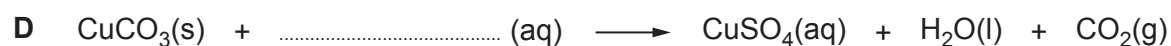
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3. (a) **A, B, C** and **D** are equations for reactions used to make salts.

(i) Complete the equation for each reaction.

[4]



(ii) Outline the **three** steps you would carry out to get a **pure** sample of silver chloride, AgCl, from the reaction mixture in equation **B**.

[2]

.....

.....

.....

.....



- (b) Sodium chloride is made when sodium hydroxide reacts with hydrochloric acid.

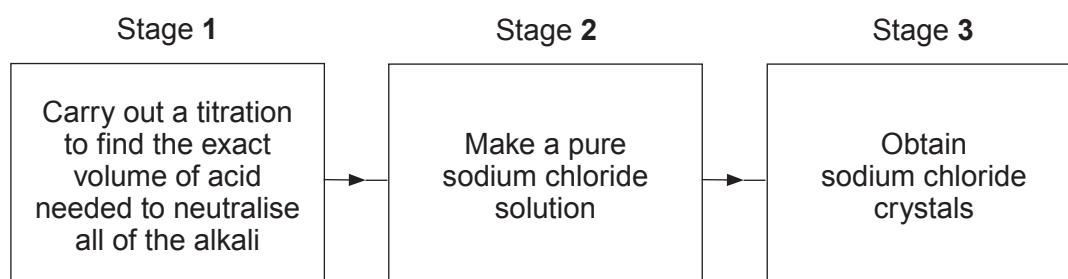


- (i) Write a balanced equation for the reaction.

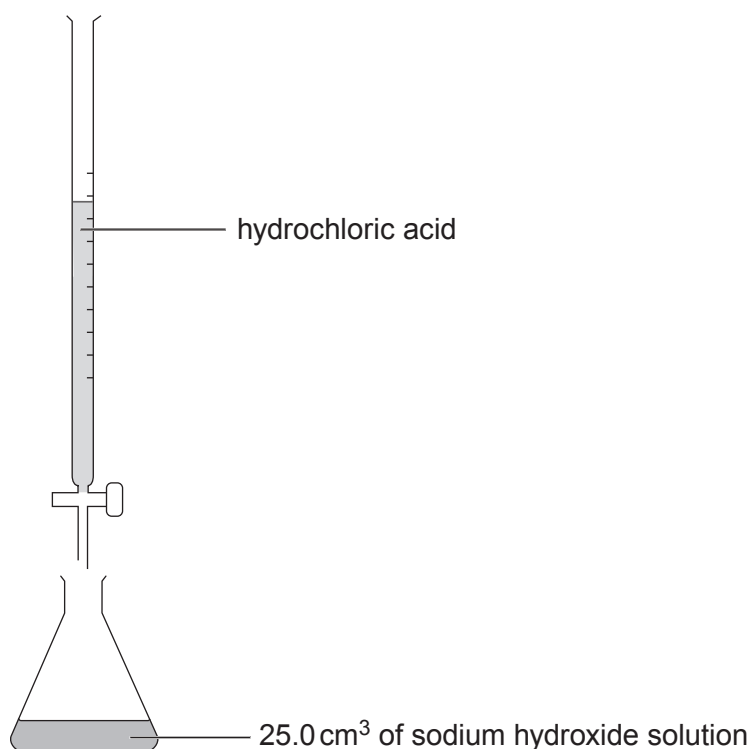
[2]

- (ii) A student was asked to prepare pure sodium chloride crystals using 25.0 cm³ of sodium hydroxide solution.

The flow diagram shows the student's plan.



- I. The student sets up the apparatus below to carry out stage 1.



Describe how the student will find the exact volume of acid needed to neutralise all of the alkali.

[2]

.....

.....

- II. The neutral solution obtained in stage 1 is not pure.

Describe what the student must do in stage 2 to make a pure sodium chloride solution.

[2]

.....

.....

.....

- III. State what the student must do in stage 3 to obtain sodium chloride crystals.

[1]

.....

.....



3. (a) The table shows information about some organic compounds.

Name	Molecular formula	Structure	Homologous series
ethanol	C_2H_5OH		alcohols
ethanoic acid	CH_3COOH	<pre> H H — C — C // \ H O O — H </pre>	
.....	C_3H_6	<pre> H H H H — C = C — C — H H H H </pre>	alkenes

(i) **Complete the table.** [3]

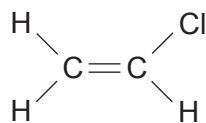
(ii) The molecular mass of an alkene is 98. Give the molecular formula of this alkene. [1]

$$A_r(C) = 12 \qquad A_r(H) = 1$$

Molecular formula

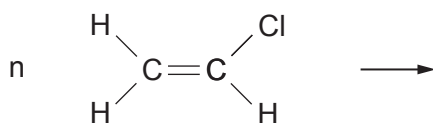


- (b) Polyvinyl chloride (PVC) is formed from the following monomer in a polymerisation reaction.



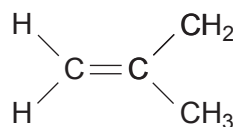
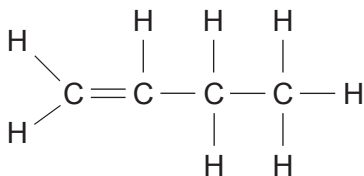
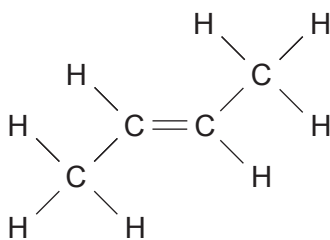
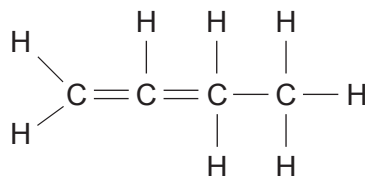
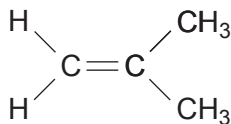
Complete the equation for the polymerisation reaction.

[2]



- (c) A student was asked to draw structures for isomers with the molecular formula C_4H_8 . She drew the following diagrams. Circle **all** the **correct** structures.

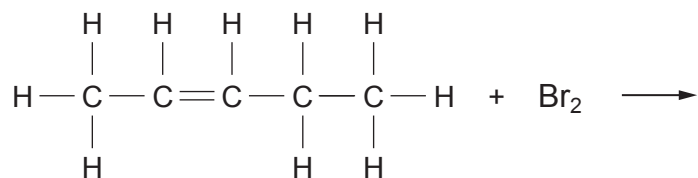
[2]



(d) Pent-2-ene reacts with bromine water in a similar manner to ethene.

Complete the equation for the reaction of pent-2-ene with bromine by drawing the structure of the product.

[1]



Examiner
only

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